Supplementary Material

Alternative structural measures and Ridge robustness

The first four figures reproduce the main structural-advantage analyses with the broker's two other saved ego-network measures, Burt's aggregate network constraint and Burt's effective size. Both are recomputed for the broker every 20 periods. Figure S5 reports the focused paired-Ridge robustness analysis discussed in the main text. As in the main text, a regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes.

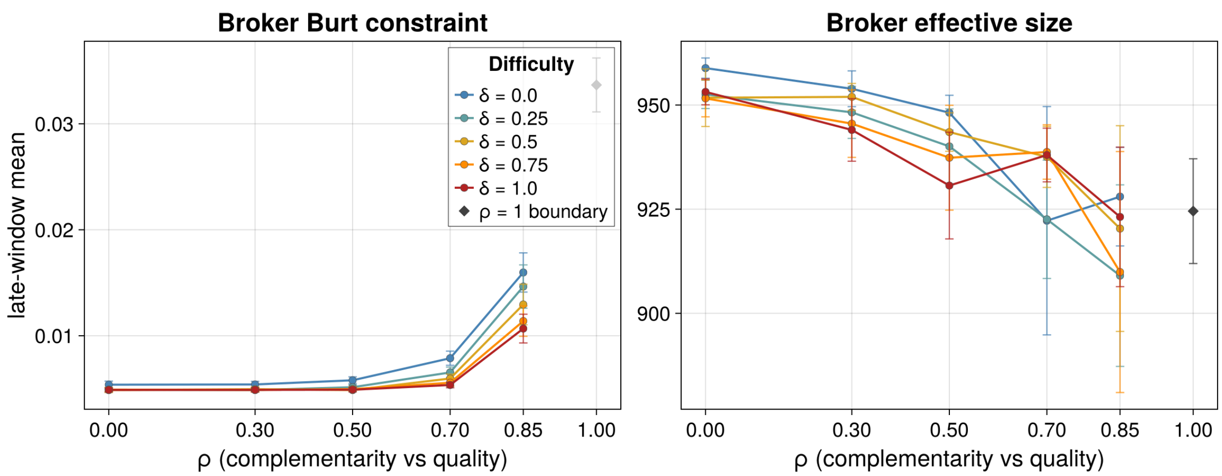

Figure S1: Broker Burt constraint (left) and broker effective size (right) across the $\rho \times \delta$ grid. Lines for each difficulty level δ span $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are late-window means over seeds and whiskers are 95% Monte Carlo intervals.

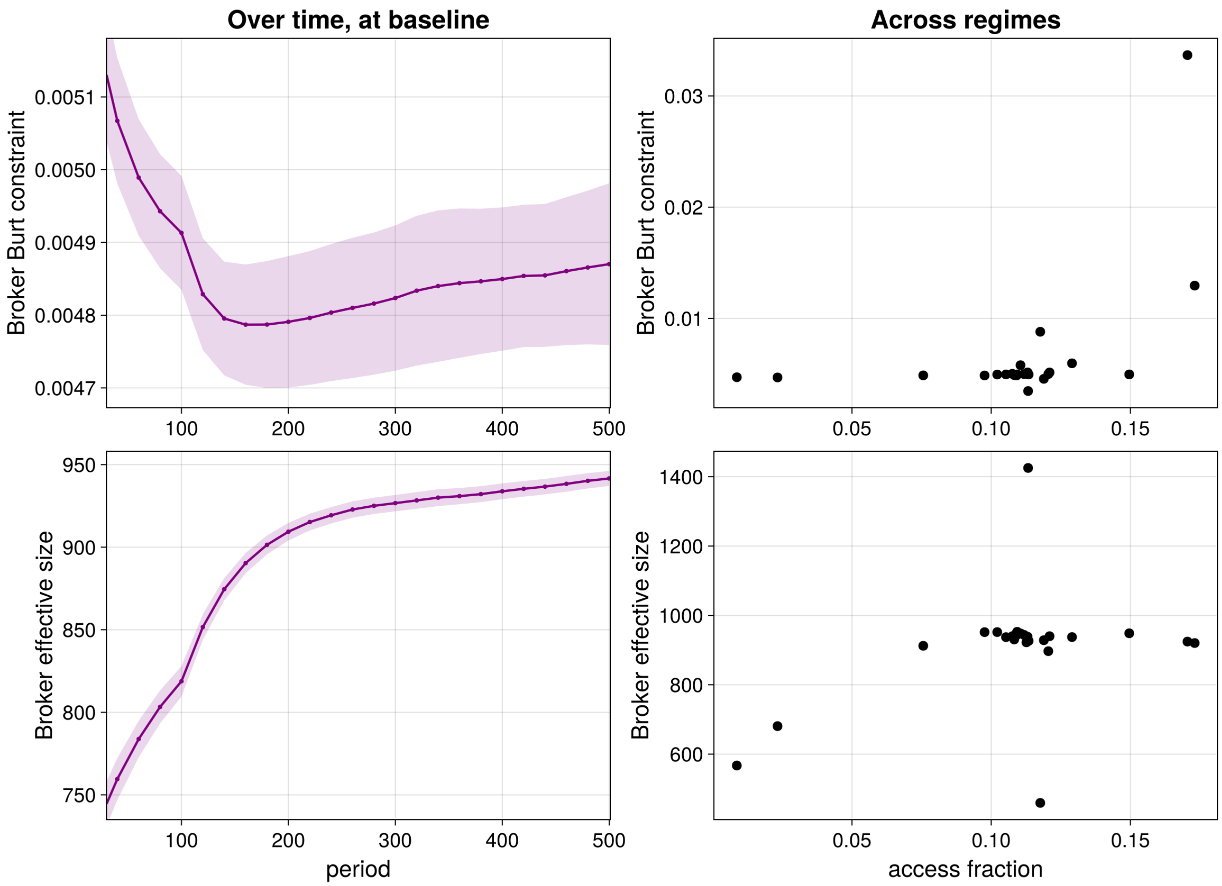

Figure S2: Broker Burt constraint (top) and broker effective size (bottom). Left: the measure over time at the baseline, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals, measured every 20 periods. Right: one dot per one-at-a-time regime, the measure against access fraction, each a late-window mean over that regime's seeds. Time starts at $t = 30$ to omit the initialization transient.

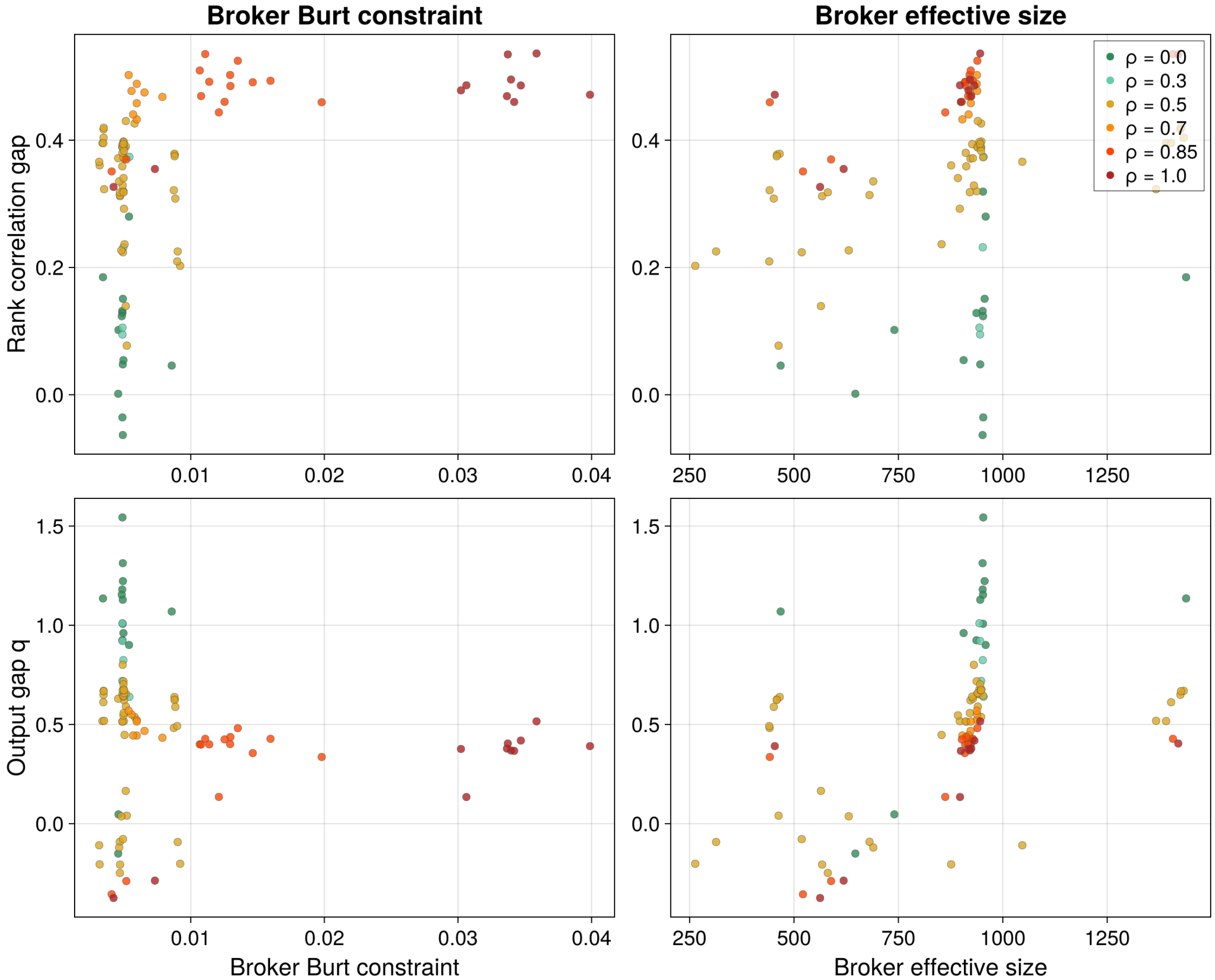

Figure S3: Rank-correlation gap (broker minus principals, top) and output gap (broker channel minus self-search, bottom) against broker Burt constraint (left) and broker effective size (right); one dot per regime, each a late-window mean over that regime's seeds, colored by the matching parameter ρ.

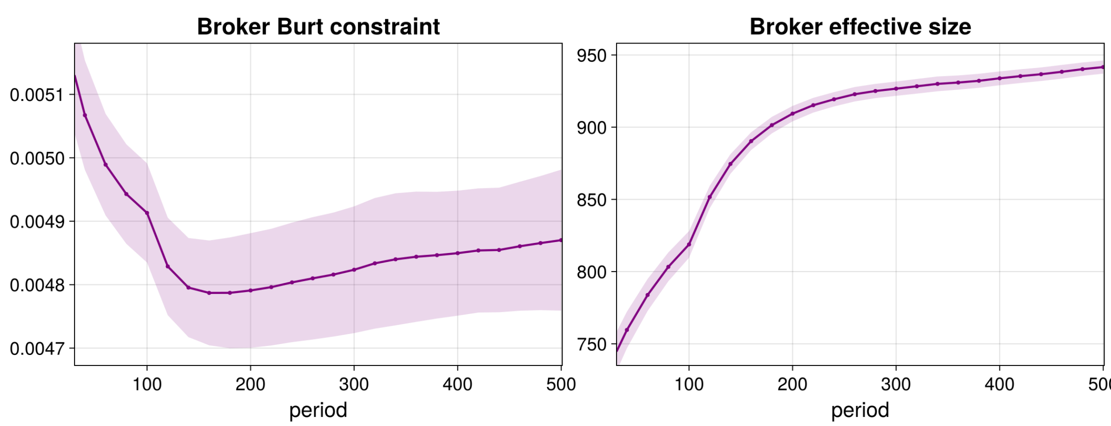

Figure S4: Broker Burt constraint (left) and broker effective size (right) over time at the baseline regime, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals, measured every 20 periods. Axes start at $t = 30$ to omit the initialization transient.

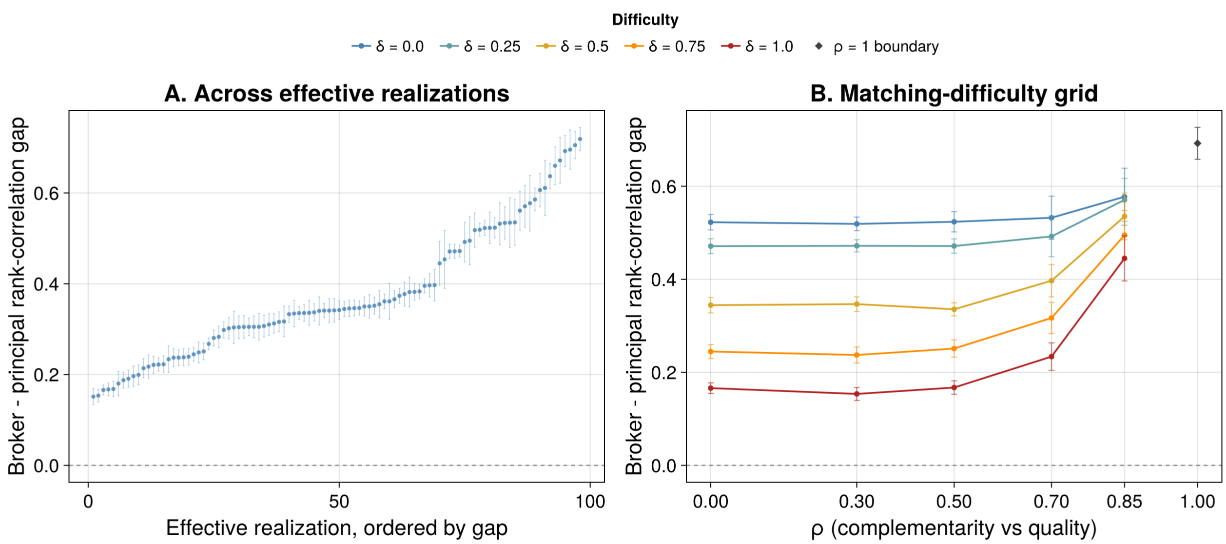

Figure S5: Broker ranking advantage under paired-Ridge learning. Panel A orders the 98 regimes by their late-period broker-minus-principal rank-correlation gap. The gap is positive in 98 regimes and has a median of 0.342; points are regime means and whiskers are 95% Monte Carlo intervals over seeds. Panel B shows the same Ridge gap across the $\rho \times \delta$ design, with one line per value of δ over $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are regime means and whiskers are 95% Monte Carlo intervals. The baseline gap is 0.335.